Urchin Foraging Rates on Kelp at Different Urchin Densities

Exploring the mutate function in RStudio through investigating urchin laboratory data from the SBC LTER.

Set Up

```
# loading in packages
library(tidyverse) # general use
library(janitor) # cleaning data
library(here) # data organization

# reading in data

# urchin density data from DiFiore et al. 2021
urchin <- read.csv(here("data","mesocosm_density_data.csv"))
```

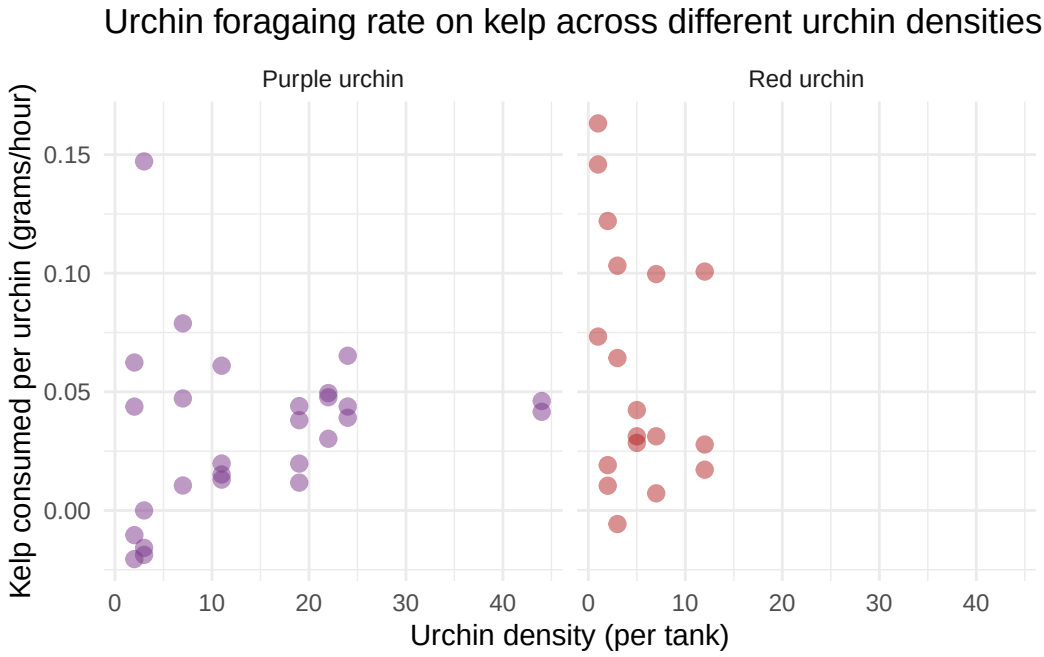
Cleaning Data & Using Mutate Function

```
# create new object urchin_clean
urchin_clean <- urchin |>
  # clean column names
  clean_names() |>
  # keeping first unique row for each unique combination
  # used to remove repetative data from each individual urchin
  distinct(trial_id, tank, .keep_all = TRUE) |>
  # filter for trials with no mortality
  filter(mortality == 0) |>
  # mutate p_r to show the different species of urchins
  rename(species = p_r) |>
```

```
  mutate(species = case_when(
    species == "p" ~ "Purple urchin",
    species == "r" ~ "Red urchin"
    )) |>
  # mutate kelp_in and kelp_out columns to the kelp foraging rate of the urchins
  mutate(
    rate_per_urchin = (kelp_in - kelp_out) / total_time / urchin_density) |>
  mutate(
    kelp_consumption_rate = (kelp_in - kelp_out) / total_time) |>
  # select for desired columns of interest
  select(urchin_density, rate_per_urchin, species, total_time, kelp_consumption_rate)
```

Visualization

```
# base plot
ggplot(data = urchin_clean,
  # x-axis urchin density, y-axis urchin consumption rate
  # color by species
       mapping = aes(x = urchin_density,
                     y = rate_per_urchin,
                     color = species)) +
  # first layer: geom_point
  geom_point(shape = 20,
             alpha = 0.5,
             size = 4) +
  # separating by species
  facet_wrap(~species) +
  # customizing colors
  scale_color_manual(values = c(
    "Purple urchin" = "mediumorchid4",
    "Red urchin" = "firebrick")) +
  # labeling axes and adding title
  labs(x = "Urchin density (per tank)",
       y = "Kelp consumed per urchin (grams/hour)",
       title = "Urchin foragaing rate on kelp across different urchin densities") +
  # adding custom theme
  theme_minimal() +
  # removing legend
  theme(legend.position = "none")
```

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       mapping = aes(x = urchin_density,
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  # first layer: geom_point
  geom_point(shape = 20,
             alpha = 0.5,
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    # separating by species
  facet_wrap(~species) +
    # customizing colors
  scale_color_manual(values = c(
    "Purple urchin" = "mediumorchid4",
    "Red urchin" = "firebrick")) +
  labs(x = "Urchin density (urchins per tank)",
       y = "Kelp consumption rate (grams/hour)",
       title = "Kelp consumption versus urchin density") +
  # adding custom theme
  theme_minimal() +
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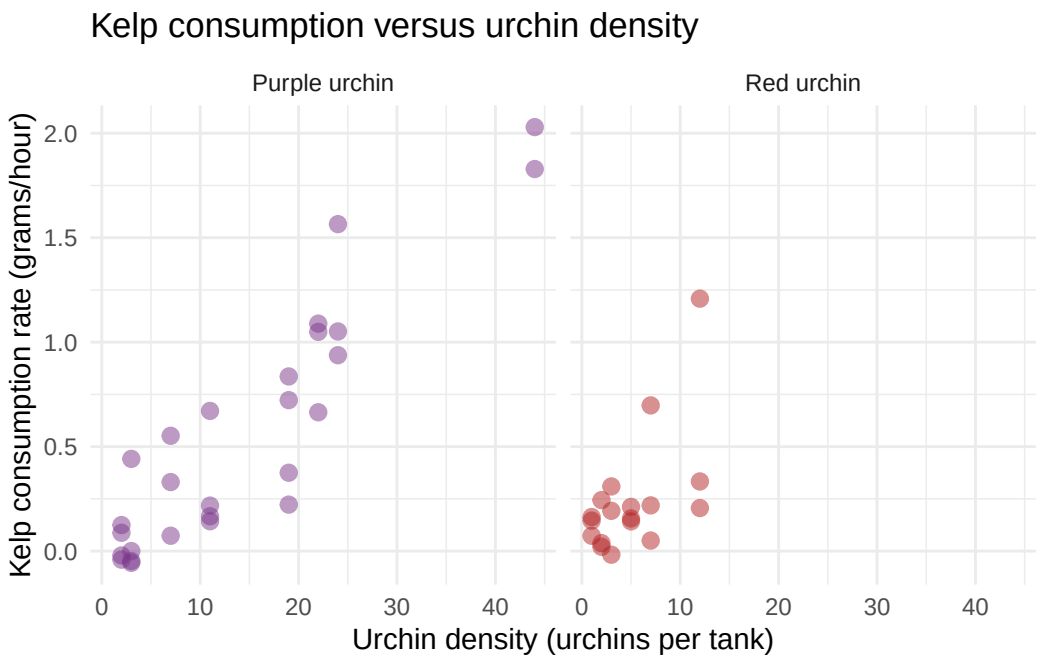
```

```
  theme(legend.position = "none")
```



Data Citation

DiFiore, B, M. Rennick, J. Curtis, D. Reed, A. Stier. 2021. SBC LTER: Sea urchin foraging rates on giant kelp ver 1. Environmental Data Initiative. https://doi.org/10.6073/pasta/6af4cc3b0e63b887baf1ae9201e1 Accessed 2026-10-16.